DSA STUDY BLUEPRINT SERIES

62. Valid Parentheses Stack

Expression Bracket Matching using Stacks

Section 08 • C++ Core Data Structures & Algorithms

Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **Valid Parentheses:** Push open brackets; match and pop on close brackets.
- Invalid if stack is empty on close check, or mismatch occurs.

Memory & Execution Blueprint

Input: "()[]{}" ⇒ Push open ⇒ Pop on close matches ⇒ Stack Empty (Valid!)

C++ Implementation Reference

Standard code layout and syntax

```
bool isValid(string s) {
    stack st;
    for (char c : s) {
        if (c=='('||c=='{'||c=='[') st.push(c);
        else {
            if (st.empty()) return false;
            char t = st.top();
            if ((c==')'&&t=='(') || (c=='}'&&t=='{') || (c==']'&&t=='[')) st.pop();
            else return false;
        }
    }
    return st.empty();
}
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Time Complexity	O(N)
Space Complexity	O(N)

Key Practices & Gotchas

- **Important:** Final stack must be completely empty to verify full expression matching.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace monotonic stack modifications for next greater element on `[4, 5, 2]`:

Active Element	Stack State (Indices)	Pop comparison	Result output
4	[0] (value 4)	N/A	Waiting for next greater
5	[1] (value 5)	5 > 4 (pop index 0)	Next Greater of 4 is 5

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Monotonic properties:** Stacks process elements strictly increasing or decreasing, optimizing nested lookups.
- **Min Stack details:** Track current minimum bounds by storing elements as encoded difference values.
- **Related LeetCode Problems:** #155 Min Stack, #84 Largest Rectangle in Histogram, #42 Trapping Rain Water.